1. Clinical Question & Core Hypothesis

Clinical Question

What molecular mechanisms determine whether immune structures translate into clinical benefit?

Working Hypothesis

In breast cancer, the prognostic benefit of high CD8+ TIL infiltration on overall survival is conditional on tumor cell apoptotic competence and metabolic/survival signaling. Specifically, tumors with high BCL2L11 (Bim) expression and low MTOR pathway activity will exhibit the strongest association between TIL density and improved OS, whereas tumors with low BCL2L11 or high MTOR activity will show attenuated or absent TIL-derived survival benefits. This interaction is further modulated by inflammatory state markers (e.g., S100A9) and immune-state indicators (e.g., CD44) as well as TGF-β–biased CD8 dynamics, which can dampen apoptotic execution and/or sustain tumor metabolism under immune pressure. The hypothesis yields testable predictions: (1) in TCGA-BRCA and similar cohorts, the TIL-OS association will be strongest in BCL2L11-high/MTOR-low tumors and weakest in BCL2L11-low/MTOR-high tumors; (2) the three-way interaction TIL × BCL2L11 × MTOR will significantly improve model fit for OS beyond any two-way interactions; (3) functional studies will show that MTOR inhibition or forced Bim upregulation sensitizes MTOR-high tumor cells to CTL-mediated killing; (4) modulation of S100A9 and CD44/TGF-β–related states will shift these interactions by altering CTL function or tumor cell susceptibility to apoptosis. Clinically, combining MTOR pathway modulation or Bim-enhancing strategies with adoptive/antigen-specific immunotherapies could convert TIL-rich but MTOR-driven tumors into OS responders.

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Anchor gene(s)	BCL2L11, MTOR
Primary endpoint	OS

2. Why This Gene — Not Others

Primary hypothesis gene(s)

BCL2L11: BCL2L11) both influence cell death and survival, yet not all TIL-rich tumors show better outcomes—why does immune presence sometimes fail to translate into improved survival? 2) Tier 1 paths used: the Bim (BCL2L11)-driven apoptotic execution axis (Bim upregulation drives tumor cell death) and the MTOR survival/metabolic signaling axis (mTOR activity can propagate pro-survival signaling that may blunt apoptosis).

MTOR: MTOR) that determines whether TILs effect tumor killing; these reagents remain important modulators for follow-up interventional analyses but would dilute the primary mechanistic question if included in the core hypothesis.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core

Secondary	S100A9	Supporting modifier from knowledge graph
Secondary	CD44	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: and TGF-β–driven CD8+ T-cell states can further modulate both CTL activity and tumor-intrinsic apoptosis, contributing to observed heterogeneity in immune response and survival even among TIL-rich tumors.

Prediction 2: retrospective analyses of TCGA-BRCA/METABRIC to model OS with TIL density, BCL2L11 and MTOR activity (as signatures or RPPA data), including three-way interaction terms and modifiers (S100A9, CD44, TGF-β signatures);

Prediction 3: in vitro co-culture models with breast cancer cell lines engineered to modulate BCL2L11 (overexpression/knockdown) and MTOR (activators/inhibitors), assessing CTL-mediated killing (Annexin V/PI, cleaved caspase-

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Gene expression does not equal protein activity; phospho-protein or functional assays are needed for confirmation.

Alternative Directions

- S100A9: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- CD44: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.